Wronskian Solutions of (Supersymmetric) Bethe Ansatz Equations: Formalism and a Bertini-Based Numerical Continuation Pipeline

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Abstract

This report develops the Q-system and Wronskian coefficient construction implemented in the `BAE_Bertini` project. We derive polynomial degrees from Young-diagram geometry, explain half-unit spectral shifts and monic QQ normalization, and describe gauge fixing, mixed discrete integration, and boundary marked points. A standard Young tableau specifies a sequence of asymptotic seeds; finite-parameter refinement and homotopy continuation carry each nontrivial growth step to the homogeneous chain. We derive the Davidenko tangent equation and show a numerical predictor–corrector step, separating equation residuals, conditioning, and tableau branch identity. Rigged configurations provide an independent combinatorial count and a string approximation, while evacuation supplies a numerical root-negation check. Two explicit examples connect the definitions to calculations: $(2, 1)$ gives an elementary quadratic, and a particular tableau of shape $(3, 2, 1)$ gives six Wronskian coefficient equations, an exact cubic, two coupled nesting levels, and a complex conjugate root pair. All QQ cells and polynomial Bethe-equation remainders of the six-box solution are verified, and an accompanying independent calculation reproduces its box-by-box growth and evacuated partner. TikZ diagrams visualize degrees, QQ cells, the numerical construction, tableau growth, and the roots. The existing saved-reference validation and performance results are reported separately from these pedagogical checks.

Contents

1 Introduction **4**
- 1.1 Physical setting . . . 4
- 1.2 Numerical strategy . . . 4
- 1.3 Scope and evidentiary status of this report . . . 4
- 1.4 How to return to this report after a long break . . . 5
- 1.5 Outline . . . 5

2 The Q-System Formalism **6**
- 2.1 Conventions and the information carried by a tableau . . . 6
- 2.2 Q-functions on the Young diagram: derive the degrees . . . 7
- 2.3 The Hirota (QQ) equation and its normalization . . . 8
- 2.4 Casoratians: determinants, gauge fixing, and polynomiality . . . 8

3 The Supersymmetric Wronskian and the Marked-Point Freedom **10**
- 3.1 A marked point is a boundary coordinate . . . 10
- 3.2 Shifted weights specify elementary degrees . . . 10
- 3.3 Discrete integration and the super-Wronskian . . . 11
- 3.4 Two coefficient bases and a practical reconstruction rule . . . 12

4 The Deformation Parameter and the Sequential Construction **12**
- 4.1 The chain inhomogeneity as a homotopy parameter . . . 12
- 4.2 The asymptotic weights, written explicitly . . . 13
- 4.3 Sequential construction: which data change at each step? . . . 13
- 4.4 Extraction of Bethe roots and the root-equation convention . . . 14

5 Rigged Configurations, the String Hypothesis, and a Flip Symmetry **15**
- 5.1 Strings and vacancy numbers . . . 15
- 5.2 Why the vacancy formula and binomial factor have this form . . . 16
- 5.3 The Kerov–Kirillov–Reshetikhin bijection . . . 16
- 5.4 A flip involution and a conjectured root symmetry . . . 17

6 Numerical Continuation: Formalism and Realization **17**
- 6.1 A concrete warm-up example . . . 17
- 6.2 Homotopy continuation and the Davidenko equation . . . 18
- 6.3 Specialization: a linear parameter homotopy in Λ . . . 18
- 6.4 A predictor–corrector step, with actual numbers . . . 19
- 6.5 Residuals, conditioning, and branch identity . . . 19
- 6.6 Numerical safeguards . . . 20
- 6.7 A path-tracking failure mode: branch jumps . . . 20

7 A Fully Worked Example: the Shape $(2,1)$ **21**

8 A Nontrivial Worked Solution: One Tableau of Shape $(3,2,1)$ **21**
- 8.1 Choose a tableau, rather than only a shape . . . 21
- 8.2 Count degrees and write every unknown . . . 22
- 8.3 Follow the branch through all nontrivial shapes . . . 23
- 8.4 Solve the selected homogeneous branch exactly . . . 24

8.5 Reconstruct the whole Q-system and check every cell . . . 25
8.6 Extract the cubic roots, with no unspecified solver step . . . 25
8.7 Strings and evacuation: what this example validates . . . 26

9 Software Realization **27**

10 Validation and Numerical Results **28**
10.1 Regression testing . . . 28
10.2 Validation of Conjecture 5.1: the flip check . . . 28
10.3 Performance profile of a single tableau . . . 29

11 Discussion **29**
11.1 Summary . . . 29
11.2 Limitations . . . 29
11.3 Outlook . . . 30

1 Introduction

1.1 Physical setting

Integrable spin chains of rank higher than one admit a *nested* Bethe ansatz: magnon excitations are organized into several coupled families of rapidities, one family per node of an auxiliary root system, and the full spectrum is obtained by solving a coupled system of Bethe ansatz equations (BAE) across all families simultaneously. For a chain of length L, the spectrum organizes into multiplets labeled by partitions λ of L (Young diagrams); the multiplicity space of a shape λ has a basis labeled by the standard Young tableaux (SYT) of that shape — fillings of the diagram's L boxes with the integers $1, \ldots, L$ that increase along every row and down every column. This report is a self-contained account of a computational pipeline that, given a single SYT, numerically constructs the corresponding Bethe roots.

Rather than attacking the Bethe equations directly in root variables, the pipeline uses the modern *Q-system* reformulation of the nested Bethe ansatz, in which the fundamental objects are polynomials $Q_{a,s}(u)$ of the spectral parameter u, attached to nodes (a, s) of a lattice built from the Young diagram, and required to satisfy a discrete bilinear functional relation (Section 2). The distinguishing feature of this project is that the Q-system is not solved by treating the coefficients of every $Q_{a,s}$ as independent unknowns; instead, a parameterization is constructed as a **Wronskian (Casoratian) determinant** of a small number of elementary building blocks, split into a *bosonic* and a *fermionic* family according to a chosen *marked point* on the diagram (Section 3). This is the origin of the name *(SUSY) Wronskian Bethe Equations* carried by the project.

1.2 Numerical strategy

Even after the Wronskian ansatz reduces the problem to finitely many polynomial equations in finitely many coefficients, those equations must still be solved numerically for each SYT. The pipeline does this by introducing a single auxiliary parameter Λ — realized physically as one chain site's inhomogeneity — and constructing a combinatorial large-Λ approximation associated with the tableau's growth sequence (Section 4). The physical, homogeneous point $\Lambda = 0$ is then reached by *numerically continuing* this solution as Λ is driven from its large starting value to zero — a textbook parameter homotopy continuation problem, which we formalize in Section 6. The tableau itself is built up one box at a time, each new box's site briefly appearing at large Λ and then being continued into the homogeneous chain, so that a single SYT computation is in fact a sequence of continuations through its nontrivial subtableaux, each seeded by the previous one's converged solution (Section 4.3).

The actual path tracking is delegated to **Bertini**, a numerical algebraic geometry package implementing adaptive multiple-precision predictor-corrector continuation, through its Python interface. The symbolic construction of the Wronskian relations, and the orchestration of many such continuations across all SYT of a Young diagram, is carried out in the Wolfram Language; the two layers communicate through an explicit polynomial system exported to a plain-text file. We describe this numerical layer formally in Section 6 and summarize its software realization in Section 9.

1.3 Scope and evidentiary status of this report

This report distinguishes four kinds of statements: definitions and algebra derived explicitly here; formulas read directly from the source; numerical evidence from saved project results; and mathematical results whose proofs belong to external literature. The equation-number comments in

the Wronskian implementation match the shifted weights, degree formulas, and Young-diagram Q-function construction of Chernyak, Leurent, and Volin [1]. The rational Q-system viewpoint and polynomiality constraints are also explained by Marboe and Volin [2]. This report derives the conventions and small examples rather than claiming to reproduce the full completeness proof.

The ordinary first-column root interpretation is used for the explicit examples. Auxiliary mixed marked points provide alternative coefficient parameterizations; they do not by themselves identify a different physical Hamiltonian. In particular, a low residual checks equations, whereas the tableau-to-branch assignment requires continuation evidence. The companion calculation in Section 8 is an independent algebraic and numerical check, distinguished from the saved Wolfram/Bertini runs.

1.4 How to return to this report after a long break

Read Section 2 first for index conventions, degree counting, and monic normalizations. Section 3 explains which coefficients are genuinely unknown and which are gauge choices. Section 4 specifies the seed and its finite-parameter refinement. Then work through Sections 7 and 8 with pencil and paper: they show the actual polynomial equations, branch selection, and root extraction. Use Section 6 to interpret numerical accuracy and branch checks. All shifts use $i/2$, all lattice polynomials are monic, and box coordinates are one-based while lattice coordinates are zero-based. These three conventions should be fixed before comparing any formulas with source code or another reference.

1.5 Outline

Section 2 introduces the Q-system and explains the Wronskian mechanism behind the Hirota/QQ relation. Section 3 constructs the supersymmetric Wronskian solution and its marked-point freedom. Section 4 introduces the deformation parameter Λ, its asymptotic solution, and the sequential construction. Section 5 develops the rigged-configuration / string-hypothesis combinatorics and the flip symmetry used for validation. Section 6 formalizes homotopy continuation and Bertini's realization of it. Section 7 checks the shape $(2, 1)$. Section 8 solves a specific tableau of shape $(3, 2, 1)$ with all determinant coefficients, growth steps, QQ cells, and roots explicit. Section 9 summarizes the software pipeline. Section 10 reports validation and performance results. Section 11 discusses scope, limitations, and outlook. Table 1 below collects the recurring notation for reference.

Symbol	Meaning
$\lambda \vdash L$, λ'	Young diagram (L boxes) labeling a multiplet, and its conjugate
SYT	standard Young tableau: a filling of λ labeling one eigenstate
u	spectral parameter (rapidity)
$Q_{a,s}(u)$	Q-function at lattice node (a, s); $M(a, s)$ its polynomial degree
$\theta_1, \ldots, \theta_L$	chain-site inhomogeneities; all zero for a homogeneous chain
$\mathrm{Wr}[f, g]$	discrete Wronskian (Casoratian) of f, g, Eq. (10)
$\mathrm{mP} = (\lambda_*, \nu_*)$	boundary coordinate selecting elementary bosonic/fermionic index sets
$Q_{A\|J}$, $\gamma_{A,J,k}$	Q-function (resp. its coefficients) for bosonic set A, fermionic set J
$c_{a,s,k}$	generic-basis monomial coefficients of $Q_{a,s}$ (Section 3.4)
Λ, Λ_0	continuation (deformation) parameter and its large starting value
$n_s^{(a)}$, J, $P_s^{(a)}$	string multiplicity, rigging, and vacancy number (Section 5)
t, $H(x, t)$	homotopy parameter and homotopy function (Section 6)

Table 1: Recurring notation.

2 The Q-System Formalism

2.1 Conventions and the information carried by a tableau

A partition $\lambda = (\lambda_1, \ldots, \lambda_\ell)$ of L is a nonincreasing list of positive row lengths with $\sum_r \lambda_r = L$. We draw it in English convention: row $r = 1$ is at the top, column $c = 1$ is at the left. A box has one-based coordinates (r, c); a lattice vertex has zero-based coordinates (a, s). Confusing these two conventions produces an off-by-one error in both the degree and asymptotic formulas. The conjugate partition is $\lambda'_c = \#\{r : \lambda_r \geq c\}$. A standard Young tableau (SYT) is a filling by $1, \ldots, L$, increasing strictly along rows and columns. Restricting a tableau T to entries at most n gives a Young diagram $\lambda^{(n)}$; hence T specifies *an order of adding boxes*, not just a final shape.

In the ordinary $\mathfrak{gl}_\ell$ interpretation, a shape specifies a symmetry representation, while different tableaux label solutions in its multiplicity space. They need not have the same energy. The degeneracy within a symmetry multiplet and the number of tableaux of one shape are therefore different notions. In this report the tableau label is attached to an asymptotic solution branch and carried to the homogeneous point by the chosen continuation procedure.

All spectral variables are expressed in units in which the elementary shift is i. We write

$$f^{[k]}(u) = f\left(u + \frac{ik}{2}\right), \qquad f^{\pm} = f^{[\pm 1]}, \qquad f^{\pm\pm} = f^{[\pm 2]}.$$

Thus $+$ means $i/2$, not i; the double shift is i. A Q-function is a polynomial whose zeros encode a family of rapidities:

$$Q(u) = \prod_{j=1}^{M} (u - u_j) = u^M + c_1 u^{M-1} + \cdots + c_M, \quad c_k = (-1)^k \sum_{j_1 < \cdots < j_k} u_{j_1} \cdots u_{j_k}. \tag{1}$$

The coefficients are symmetric in the roots, so permuting roots does not create a new coefficient solution. A real coefficient vector can encode complex conjugate roots; reality of coefficients never implies reality of all Bethe roots. For $M = 0$ we set $Q = 1$, with an empty root set.

2.2 Q-functions on the Young diagram: derive the degrees

At vertex (a, s), delete the first a rows and first s columns. The number of surviving boxes is

$$M(a,s) = \#\{(r,c) \in \lambda : r > a,\ c > s\} = \sum_{r>a} \max(\lambda_r - s, 0). \tag{2}$$

This counting definition works everywhere. At the vertices used by the QQ relations, the overlap of the removed rows and columns is a complete $a \times s$ rectangle, so inclusion–exclusion gives the code's formula

$$M(a,s) = L + as - \sum_{r=1}^{a} \lambda_r - \sum_{c=1}^{s} \lambda'_c. \tag{3}$$

Do not extrapolate the rectangle formula to arbitrary vertices far outside the diagram: its overlap assumption can fail there. Boundary vertices with no surviving boxes have $M = 0$ and $Q_{a,s} = 1$. The distinguished vacuum polynomial is fixed data,

$$Q_{0,0}(u) = \prod_{n=1}^{L} (u - \theta_n). \tag{4}$$

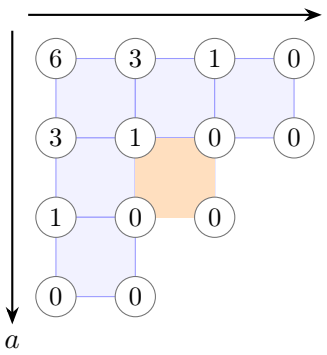
The θ_n are site inhomogeneities, not unknown Bethe roots. For the homogeneous chain, $Q_{0,0} = u^L$.

The hook length of box (r, c) is $h_{r,c} = \lambda_r - c + \lambda'_c - r + 1$: the box itself, the boxes to its right, and those below it. Removing one more row and column at a QQ cell removes precisely this hook. Consequently,

$$M(a,s) - M(a+1,s+1) = h_{a+1,s+1}, \tag{5}$$

$$M(a,s) + M(a+1,s+1) - 1 = M(a+1,s) + M(a,s+1). \tag{6}$$

The second equality follows by applying inclusion–exclusion to the four southeast regions. These identities explain both the degree and the normalization of the QQ equation.



The numbers at vertices are $M(a, s)$.

At $(1, 1)$, only the orange box survives: $M(1, 1) = 1$.

First-column degrees: $6, 3, 1, 0$.
Physical nesting levels: $Q_{1,0}$ and $Q_{2,0}$.

Figure 1: The shape $(3, 2, 1)$ and its Q-function degrees. Vertex coordinates count removed rows and columns; they are not box coordinates.

Example 2.1 (A first check: $\lambda = (2, 1)$)**.** *Here $L = 3$, $\lambda' = (2, 1)$, and $M(0, 0) = 3$, $M(1, 0) = M(0, 1) = 1$; all other relevant vertices have degree zero. There are two tableaux,*

$$T_A = \begin{smallmatrix} 1 & 2 \\ 3 \end{smallmatrix}, \qquad T_B = \begin{smallmatrix} 1 & 3 \\ 2 \end{smallmatrix},$$

with growth sequences $(1) \to (2) \to (2, 1)$ and $(1) \to (1, 1) \to (2, 1)$. The same final polynomial system can therefore be approached from two different seeds. There is one ordinary nesting level, containing the single zero of $Q_{1,0}$.

2.3 The Hirota (QQ) equation and its normalization

The monic Q-functions must obey, for each box $(a+1, s+1)$,

$$Q^{+}_{a,s}Q^{-}_{a+1,s+1} - Q^{-}_{a,s}Q^{+}_{a+1,s+1} = i\big(M(a,s) - M(a+1,s+1)\big)Q_{a+1,s}Q_{a,s+1}. \tag{7}$$

This is a polynomial functional identity in u. The underlying rational Bethe equations are also algebraic after denominators are cleared; introducing Q-functions changes their organization and polynomiality constraints, rather than changing a transcendental problem into an algebraic one.

To see why the prefactor is necessary, let $f = u^p + \cdots$ and $g = u^q + \cdots$. Since $(u \pm i/2)^p = u^p \pm ipu^{p-1}/2 + \cdots$,

$$f^{+}g^{-} - f^{-}g^{+} = i(p-q)u^{p+q-1} + \cdots . \tag{8}$$

The leading u^{p+q} terms cancel. Equation (6) fixes the degree of the right side, and Eq. (5) fixes its leading coefficient. A raw determinant is generally not monic; dropping the factor $i(p-q)$ changes the coefficient equations.

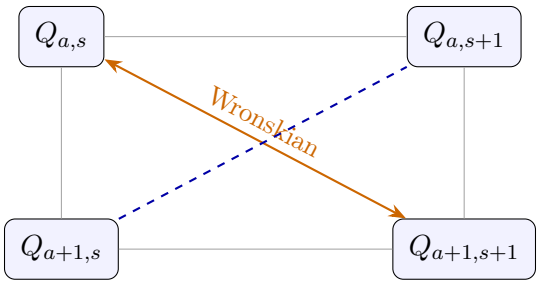


Figure 2: One QQ cell. Take the discrete Wronskian of the northwest and southeast polynomials; it equals the product of the other two, multiplied by i times the hook length. The shifts act on u, not on (a, s).

How an identity in u becomes equations in coefficients

Write $Q_{a,s} = u^{M(a,s)} + \sum_{k=1}^{M(a,s)} c_{a,s,k}u^{M(a,s)-k}$. Substitute this ansatz into Eq. (7), expand every shift by the binomial theorem, and require each coefficient of the resulting polynomial in u to vanish. This is an identity for all u, not an equation tested at one arbitrarily chosen spectral value.

For $(2, 1)$ at homogeneity, set $Q_{1,0} = u+x$, $Q_{0,1} = u+y$, $Q_{1,1} = 1$. The only nontrivial QQ relation is

$$(u + i/2)^3 - (u - i/2)^3 = 3i(u + x)(u + y).$$

Dividing by $3i$ gives $u^2 - 1/12 = u^2 + (x + y)u + xy$, so

$$x + y = 0, \qquad xy = -\frac{1}{12}, \qquad x = \pm\frac{1}{2\sqrt{3}}, \quad y = -x. \tag{9}$$

The physical root is $-x$, because $Q_{1,0} = u+x$. This sign is a useful check when comparing coefficient and root files.

2.4 Casoratians: determinants, gauge fixing, and polynomiality

Define the two-function discrete Wronskian by

$$\mathrm{Wr}[f, g] = f^{+}g^{-} - f^{-}g^{+}. \tag{10}$$

For k functions, its centered Casoratian is

$$\operatorname{Wr}[f_1,\dots,f_k](u)=\det_{1\le r,j\le k} f_r\left(u+\frac{i}{2}(k+1-2j)\right). \tag{11}$$

For example, the shifts are $(i/2,-i/2)$ for $k=2$ and $(i,0,-i)$ for $k=3$. A Casoratian uses shifted values; an ordinary Wronskian uses derivatives. In the small-shift expansion, subtracting nearly identical columns explains why the leading degree drops by $k(k-1)/2$. For monic polynomials of distinct descending degrees $d_1>\cdots>d_k$,

$$\deg \operatorname{Wr}=\sum_r d_r-\frac{k(k-1)}{2},\qquad \operatorname{lc}_u \operatorname{Wr}=i^{k(k-1)/2}\prod_{r<t}(d_r-d_t). \tag{12}$$

One can derive the coefficient by expanding each shifted polynomial, then taking the determinant of the first k nonvanishing powers of the shift; the two Vandermonde factors give the product of degree differences and the power of i. For a nonzero polynomial P, $\operatorname{monic} P = P/\operatorname{lc}_u P$. All normalized determinants below mean division by this nonzero leading coefficient.

For the purely bosonic marked point $(\ell,0)$, use elementary polynomials B_r of degrees

$$d_r=\lambda_r+\ell-r,\qquad Q_{a,0}=\operatorname{monic}\operatorname{Wr}[B_{a+1},\dots,B_\ell]. \tag{13}$$

Equation (12) gives $\deg Q_{a,0}=\sum_{r>a}\lambda_r=M(a,0)$ as required. Adding a lower-degree B_t to B_r with $t>r$ leaves all these nested determinants unchanged. Fix this freedom by making B_r monic and setting its coefficients of powers u^{d_t}, $t>r$, to zero. This is a choice of basis, not an extra Bethe equation. It leaves $d_r-(\ell-r)=\lambda_r$ free coefficients in B_r, hence exactly L in total. Without gauge fixing, equivalent elementary bases introduce redundant unknowns and may make the numerical Jacobian singular.

Determinant identities, rather than an assumed differential equation, provide the algebraic mechanism. The Desnanot–Jacobi identity

$$\det D\,\det D^{ij}_{ij}=\det D^i_i\,\det D^j_j-\det D^j_i\,\det D^i_j$$

relates complementary minors of one matrix. Here D^j_i deletes row i and column j, and D^{ij}_{ij} deletes both indicated rows and columns; deleting extreme shift columns produces neighboring shifted Casoratians. After normalizing minors, these identities yield QQ relations. The precise graded construction and its completeness are treated in [1]; here we use its formulas as implemented and verify all QQ cells explicitly in the six-box example.

In the index-set notation of the next section, the code recursively uses

$$Q_{A_0\cup\{a,b\}|J}=\frac{\operatorname{Wr}[Q_{A_0\cup\{a\}|J},Q_{A_0\cup\{b\}|J}]}{Q_{A_0|J}}, \tag{14}$$

$$Q_{A|J_0\cup\{i,j\}}=\frac{\operatorname{Wr}[Q_{A|J_0\cup\{i\}},Q_{A|J_0\cup\{j\}}]}{Q_{A|J_0}}, \tag{15}$$

with conventions for raw normalizations fixed consistently within that basis. A quotient of polynomials need not be polynomial: if the numerator is $N(u)$ and the denominator is $D(u)$, Euclidean division $N=DS+R$, $\deg R<\deg D$, requires $R\equiv 0$. Cancelling a nonzero remainder, or checking a quotient only away from its poles, loses the polynomiality condition. This distinction matters particularly when some shifted Q-functions share zeros [2].

For intuition only, two solutions of $X(u+i)+X(u-i)=V(u)X(u)$ obey

$$X(u+i)+X(u-i)=V(u)X(u), \qquad X\in\{f,g\}, \tag{16}$$

so

$$\mathrm{Wr}[f,g]^{+}-\mathrm{Wr}[f,g]^{-}=g(f^{++}+f^{--})-f(g^{++}+g^{--})=0. \tag{17}$$

This illustrates cancellation under shifts. It does not prove the nonzero-right-hand-side QQ equation, nor does it specify an additional linear equation that the implementation solves.

3 The Supersymmetric Wronskian and the Marked-Point Freedom

3.1 A marked point is a boundary coordinate

Write $\mathrm{mP}=(m,n)=(\lambda_*,\nu_*)$ for a lattice vertex on the southeast boundary of the Young diagram, so $M(m,n)=0$. The m elementary bosonic labels are $1,\dots,m$; the n elementary fermionic labels are $1,\dots,n$. These are auxiliary index sets. They do not partition the rows of the diagram: there is no general requirement $m+n=\ell$. The shape must fit the corresponding fat hook, $\lambda_{m+1}\le n$, and the marked vertex lies on its boundary. A monotone path from $(0,0)$ to the marked point specifies an ordering of the bosonic and fermionic removals, hence a nesting/grading choice. The endpoint and the path are distinct data.

The code enumerates boundary vertices as

$$\mathcal{E}(\lambda)=\{(0,\lambda_1)\}\cup\bigcup_{a=1}^{\ell-1}\{(a,s):\lambda_{a+1}\le s\le\lambda_a\}\cup\{(\ell,s):0\le s\le\lambda_\ell\}. \tag{18}$$

The displayed rectangular determinant below applies for $m\ge n$; random mixed-point selectors in the current workflow restrict further to $m>n$. The default selector is the deterministic purely bosonic point $(\ell,0)$. Thus an enumeration of efficient paths is not a claim that every production run chooses a globally minimal one.

Example 3.1 (Marked points for $(2,1)$)**.** *The boundary points are* $(0,2),(1,1),(1,2),(2,0),(2,1)$. *The purely bosonic point* $(2,0)$ *and the balanced point* $(1,1)$ *both parameterize this shape, although the current random selectors exclude the balanced point. The example* $(1,1)$ *below is still useful for understanding the mixed construction. The point* $(0,2)$ *requires a fermionic counterpart of the determinant formula, which we do not pretend is covered by a matrix with a negative number of bosonic columns.*

3.2 Shifted weights specify elementary degrees

For a chosen marked point, the code's `lambdaMarked` and `nuMarked` give

$$\widehat{\lambda}_p=\lambda_p-p-n+m \quad (1\le p\le m), \qquad \widehat{\nu}_q=\lambda'_q-q-m+n \quad (1\le q\le n). \tag{19}$$

These are degrees, not row lengths. For bosonic and fermionic sets $A\subseteq\{1,\dots,m\}$ and $J\subseteq\{1,\dots,n\}$, respectively,

$$M_{A|J}=\sum_{p\in A}\widehat{\lambda}_p+\sum_{q\in J}\widehat{\nu}_q-\frac{|A|(|A|-1)}{2}-\frac{|J|(|J|-1)}{2}+|A||J|. \tag{20}$$

In particular, $\deg Q_{\{p\}|\emptyset} = \widehat{\lambda}_p$, $\deg Q_{\emptyset|\{q\}} = \widehat{\nu}_q$, and $\deg Q_{\{p\}|\{q\}} = \widehat{\lambda}_p + \widehat{\nu}_q + 1$. Inside the marked rectangle the distinguished lattice polynomial is

$$Q_{a,s} = \operatorname{monic} Q_{\{a+1,\dots,m\}|\{s+1,\dots,n\}}. \tag{21}$$

The complementary index convention explains why the vacuum has all indices and the marked boundary vertex has none. Vertices beyond this rectangle are reconstructed by QQ relations and boundary data.

For $(3,2,1)$ at $(3,0)$, the bosonic degrees are $(5,3,1)$. At $(2,1)$ they are $(3,1)$ and the fermionic degree is 1. The latter point has $m+n=3$, whereas the also-admissible point $(3,1)$ has $m+n=4$. These concrete checks rule out interpreting a marked point as a fixed split of three rows.

3.3 Discrete integration and the super-Wronskian

Let $B_p = Q_{\{p\}|\emptyset}$ and $F_q = Q_{\emptyset|\{q\}}$ be monic polynomials of the degrees just specified. Their elementary mixed block $R_{pq} = Q_{\{p\}|\{q\}}$ satisfies

$$R^+_{pq} - R^-_{pq} = B_p F_q. \tag{22}$$

The finite-difference operator lowers degree by one: $\Delta u^d = (u+i/2)^d - (u-i/2)^d = id\,u^{d-1} + \cdots$. Hence, with $d = \widehat{\lambda}_p + \widehat{\nu}_q + 1$, the leading coefficient of R_{pq} is $1/(id) = -i/d$. Match powers in descending order to determine its other nonconstant coefficients. Its constant κ_{pq} is free because $\Delta\kappa_{pq} = 0$. This is the discrete counterpart of a constant of integration, and must remain an unknown when matching the vacuum.

The source uses a real coefficient representation and divides the finite-difference coefficient list by its leading coefficient before solving. Thus raw factors of i and stored coefficients must not be identified blindly. For example $\widetilde{R} = R/i$ has real leading coefficient $-1/d$, and $\Delta\widetilde{R} = -iB_pF_q$; normalizing its leading coefficient recovers the same triangular algebra. Below, R denotes the raw convention of Eq. (22); physical lattice polynomials are always normalized to monic form.

For $m \geq n$, form

$$\mathcal{F}_{p,q}(u) = R_{pq}\left(u + \frac{i}{2}(m-n)\right), \qquad 1 \leq p \leq m, \quad 1 \leq q \leq n, \tag{23}$$

$$\mathcal{B}_{p,j}(u) = B_p\left(u + \frac{i}{2}(m-n-2j+1)\right), \qquad 1 \leq p \leq m, \quad 1 \leq j \leq m-n. \tag{24}$$

The first block has n columns and the second has $m-n$, so the joined matrix is square, of size m. Define the raw super-Wronskian

$$\mathcal{W}(u) = (-1)^{n(m-n)} \det[\mathcal{F} \mid \mathcal{B}](u). \tag{25}$$

The actual equations passed to the solver are

$$\operatorname{monic} \mathcal{W}(u) = \prod_{j=1}^{L} (u - \theta_j), \tag{26}$$

interpreted coefficient by coefficient. The raw determinant and the monic vacuum are different objects until this normalization is applied. At $n=0$, Eq. (25) is precisely the ordinary Casoratian of Eq. (11).

Example 3.2 (Perform the mixed integration for $(2,1)$)**.** *Let $(m,n)=(1,1)$. Both shifted weights equal 1, so $B=u+b$, $F=u+f$. Since $\Delta u=i$, $\Delta u^2=2iu$, and $\Delta u^3=3iu^2-i/4$, direct integration gives*

$$R=-i\left[\frac{u^3}{3}+\frac{b+f}{2}u^2+\left(bf+\frac{1}{12}\right)u\right]+\kappa.$$

The determinant is just R, with no residual shift. Divide by its leading coefficient $-i/3$ to get

$$\operatorname{monic} R=u^3+\frac{3}{2}(b+f)u^2+\left(3bf+\frac{1}{4}\right)u+3i\kappa.$$

Matching u^3 requires $b+f=0$, $bf=-1/12$, and $\kappa=0$. Moreover $Q_{1,0}=F=u+f$ and $Q_{0,1}=B=u+b$ by Eq. (21). This reproduces Eq. (9) using three elementary coefficients, including the integration constant. The mixed notation has not changed the physical solution.

3.4 Two coefficient bases and a practical reconstruction rule

The generic basis contains $c_{a,s,k}$ for the monic polynomials at lattice vertices. The Wronskian basis contains the remaining elementary coefficients, denoted collectively by γ (the source's `SUSYc`), after degree and gauge constraints. The latter basis can have substantially fewer unknowns. Neither u nor Λ belongs to this unknown vector: u is eliminated by coefficient matching and Λ is an external deformation parameter.

For the purely bosonic construction, calculate nested minors Eq. (13), normalize them, and read off their coefficients. This gives c as polynomial expressions in γ. For a general marked point, use Eq. (21), the recursive reductions, and QQ propagation beyond the marked rectangle. Always check that every division has zero remainder. The reverse conversion used to prepare numerical starts selects coefficient equations containing exactly one as-yet-unknown variable and only previously known quantities otherwise. Substitution then continues in triangular order. Failure to find such an equation is a failure of that elimination route; it is not proof that the physical system has a continuous family of solutions.

A useful minimal diagnostic is: identify the unknowns, count the independent equations after gauge fixing, check all expected degrees, and check every polynomial remainder. These operations establish what system is being solved before numerical precision enters.

4 The Deformation Parameter and the Sequential Construction

4.1 The chain inhomogeneity as a homotopy parameter

Suppose $T^{(n-1)}$ has already been solved at homogeneity. Introduce the next site by taking $\theta_n=\Lambda$ and $\theta_j=0$ for $j<n$:

$$Q^{(n)}_{0,0}(u;\Lambda)=u^{n-1}(u-\Lambda). \tag{27}$$

Thus the target is $\Lambda=0$, while a large positive Λ gives a controlled seed associated with the newly added box. The old solution supplies initial data; the current n-site coefficients are allowed to move during continuation. They are not held fixed throughout the solve.

4.2 The asymptotic weights, written explicitly

Let entry n occupy box (r_n, c_n), and evaluate row and column lengths in the *current* subdiagram $\lambda^{(n)}$. For a vertex northwest of this box, $a < r_n$ and $s < c_n$, define

$$N_{a,s}^{(n)} = \prod_{j=1}^{a} \frac{\lambda_j^{(n)} + r_n - c_n - j}{\lambda_j^{(n)} + r_n - c_n - j + 1} \prod_{k=1}^{s} \frac{(\lambda^{(n)})'_k + c_n - r_n - k}{(\lambda^{(n)})'_k + c_n - r_n - k + 1}, \tag{28}$$

$$q_{a,s}^{\mathrm{lead},(n)} = N_{a,s}^{(n)} \Lambda. \tag{29}$$

Empty products equal 1, so $N_{0,0}^{(n)} = 1$, consistent with the new vacuum root Λ. Only vertices northwest of the new box gain one root. The factors are ratios $(h-1)/h$ of current hook lengths in the new box's column and row. This is the formula implemented in `initThetaRep`, using the current subtableau rather than the final shape. In particular, using final-shape hooks at an earlier growth step can select the wrong branch.

For fully separated inhomogeneities, each surviving box contributes one leading root $N_{a,s}^{(n)}\theta_n$. The sequential scheme instead retains the already converged finite roots and appends the new large root. Its initial polynomial is

$$Q_{a,s}^{(n),\mathrm{seed}}(u;\Lambda) = \begin{cases} (u - N_{a,s}^{(n)}\Lambda) Q_{a,s}^{(n-1)}(u;0), & a < r_n,\ s < c_n, \\ Q_{a,s}^{(n-1)}(u;0), & \text{otherwise.} \end{cases} \tag{30}$$

Newly relevant old boundary polynomials are interpreted as 1. Equation (30) is leading asymptotic initial data, not an exact finite-Λ solution in general. The source refines it by minimizing QQ residuals at large finite values, converts the result to Wronskian coefficients, and solves the Wronskian system before handing a start point to the tracker. Only a seed with exactly vanishing residual qualifies for the source's small-tableau shortcut that skips this minimization.

Example 4.1 (An exact check of the large-Λ distinction)**.** *For $T_A = \{\{1,2\},\{3\}\}$, the new box is $(2,1)$ and $N_{1,0}^{(3)} = 2/3$. The seed root is $2\Lambda/3$. Yet the exact QQ relation at length three gives*

$$Q_{1,0}Q_{0,1} = \frac{\Delta[u^2(u-\Lambda)]}{3i} = u^2 - \frac{2\Lambda}{3}u - \frac{1}{12}.$$

The root on the large positive branch is

$$r(\Lambda) = \frac{\Lambda + \sqrt{\Lambda^2 + 3/4}}{3} = \frac{2\Lambda}{3} + \frac{1}{8\Lambda} + O(\Lambda^{-3}). \tag{31}$$

For positive real Λ, choose the positive square root continuously. At homogeneity $r(0) = 1/(2\sqrt{3})$. This proves directly that the leading seed omits a finite correction; it also fixes the tableau-to-sign assignment for this chosen path. The other branch tends to zero at large Λ and reaches the negative root at $\Lambda = 0$.

4.3 Sequential construction: which data change at each step?

The tableau defines the chain $\emptyset = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(L)}$. For each nontrivial step:

1. Build the current diagram's degree data and gauge-fixed Wronskian system $F_n(\gamma, \Lambda) = 0$.
2. Calculate Eq. (30) using the previous homogeneous solution and the newly added box.

3. Refine at a large finite Λ_0 until the current system has an acceptable start residual. Reconstruct Wronskian coefficients.

4. Track that one solution branch to $\Lambda = 0$, allowing all current unknown coefficients to change, and validate the endpoint.

5. Recover generic coefficients and use them as the next step's input.

A tableau whose initial entries all lie in the first row has an initial trivial segment. The implementation begins at the first nontrivial subtableau rather than necessarily running $L-1$ nontrivial tracker calls. A new row can also change the number of nesting levels and the size of the coefficient vector; this is why consecutive steps rebuild the system.

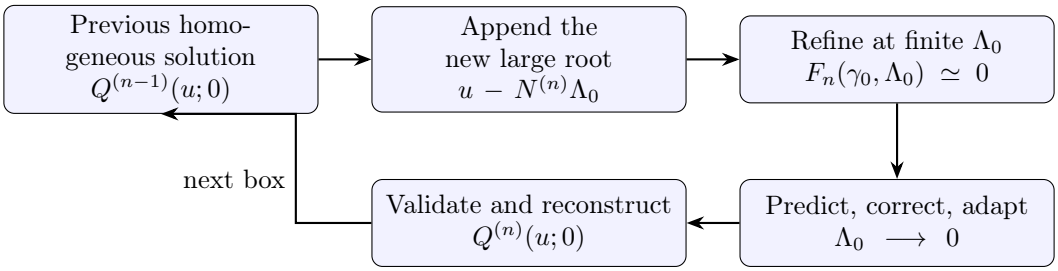


Figure 3: One growth step. The large-root formula prepares an initial approximation; refinement and continuation solve the finite system.

4.4 Extraction of Bethe roots and the root-equation convention

At the homogeneous endpoint, the ordinary first-column nesting uses

$$Q_{w,0}(u) = 0, \qquad w = 1, \dots, \ell - 1. \tag{32}$$

There are $\ell - 1$ nesting levels, with $M_w = M(w,0) = \sum_{r>w} \lambda_r$ roots at level w. The file format preserves these levels as separate lists.

For comparison with ordinary, untwisted type-A nested BAE, write $Q_w = Q_{w,0}$, $Q_0 = \prod(u - \theta_n)$ and $Q_\ell = 1$. For a simple root $u_j^{(w)}$ with nonvanishing relevant denominators,

$$\frac{Q^+_{w-1}(u_j^{(w)})}{Q^-_{w-1}(u_j^{(w)})} \frac{Q^+_{w+1}(u_j^{(w)})}{Q^-_{w+1}(u_j^{(w)})} = -\frac{Q^{++}_w(u_j^{(w)})}{Q^{--}_w(u_j^{(w)})}. \tag{33}$$

The minus sign compensates for the self-factor $(u_j - u_j + i)/(u_j - u_j - i) = -1$ inside the ratio on the right. Expanding the products therefore gives the usual scattering factors with shifts i at the same level and $i/2$ at neighboring levels. Only the first level has the chain-driving factor, since Q_0 carries inhomogeneities and $Q_\ell = 1$ contributes none.

A useful explanation of this equation is pole cancellation: a Baxter-type expression

$$\mathcal{T}_w(u) = \frac{Q^+_{w-1} Q^+_{w+1} Q^{--}_w + Q^-_{w-1} Q^-_{w+1} Q^{++}_w}{Q_w}$$

must have zero numerator at a simple zero of its denominator. Dividing by its nonzero shifted factors gives Eq. (33). This is a local check on a nesting path; polynomiality of the full Q-system

supplies additional global conditions. At singular roots, check polynomial identities or a regularized limit instead of dividing by zero.

For example, the conditioning of a simple zero follows by perturbing $Q(r) = 0$: $\delta r \simeq -\delta Q(r)/Q'(r)$. A small coefficient error can thus create a larger root error near a multiple zero. Track coefficients, but validate both coefficients and roots; their numerical accuracies need not coincide.

5 Rigged Configurations, the String Hypothesis, and a Flip Symmetry

5.1 Strings and vacancy numbers

Independently of the Wronskian construction above, the classical *string hypothesis* conjectures that solutions of the (ordinary, non-super) nested Bethe equations organize into **strings**: for each node a of the auxiliary root system, a collection of complex roots sharing a common real center v and equally spaced along the imaginary direction,

$$u_k \;=\; v - \frac{i}{2}(s - 1 - 2k), \qquad k = 0, 1, \ldots, s-1, \tag{34}$$

for a string of length s. A configuration of strings is encoded combinatorially by a **rigged configuration**: for each node a and each string length s, a nonnegative integer multiplicity $n_s^{(a)}$ together with, for every one of those $n_s^{(a)}$ strings, an integer **rigging** $J \in \{0, 1, \ldots, P_s^{(a)}\}$ bounded by the node's **vacancy number**

$$P_s^{(a)} \;=\; \delta_{a,1}\, L \;-\; \sum_b A_{ab} \sum_{s'} \min(s, s')\, n_{s'}^{(b)}, \tag{35}$$

where A_{ab} is the Cartan matrix of the auxiliary (type-A_{n-1}, in the cases implemented here) root system. Equation (35) is the standard counting-function/vacancy formula of the string hypothesis: $\delta_{a,1}L$ is the leading contribution of the physical chain to the first node, and the Cartan-matrix contraction against the "string-string scattering kernel" $\min(s, s')$ subtracts, for every other string present at neighboring nodes, the number of momentum slots it excludes at node a. A rigging J for a given string is then, in the usual interpretation, a choice of one of the $P_s^{(a)} + 1$ admissible mode numbers left available after this exclusion — exactly analogous to filling single-particle levels subject to a generalized Pauli principle. The number of admissible riggings for a fixed string-length content is therefore

$$\prod_{a,s} \binom{P_s^{(a)} + n_s^{(a)}}{n_s^{(a)}}, \tag{36}$$

a product of binomial coefficients. Summing over admissible string contents with the prescribed root numbers recovers the combinatorial count of standard Young tableaux; the fixed-content factor alone does not in general equal the full tableau count.

Example 5.1 (Vacancy numbers for $\lambda = (2,1)$)**.** *The shape* $(2,1)$ *of Example 2.1 has a single nesting level (Eq. (32)), so its auxiliary root system has a single node* $a = 1$*, with Cartan entry* $A_{11} = 2$*. Consider the string content with one string of length* $s = 1$ *and nothing else,* $n_1^{(1)} = 1$*: by (35),*

$$P_1^{(1)} \;=\; \delta_{1,1} \cdot 3 \;-\; A_{11} \cdot \min(1,1) \cdot n_1^{(1)} \;=\; 3 - 2 \cdot 1 \cdot 1 \;=\; 1,$$

so the single string's rigging can be $J = 0$ *or* $J = 1$*: exactly* $\binom{P+n}{n} = \binom{1+1}{1} = 2$ *admissible riggings, matching the two standard Young tableaux* T_A, T_B *of that shape one for one. This is the smallest*

nontrivial instance of the general fact (36): this particular shape has only one admissible string content, so its fixed-content rigging count already equals its full tableau count.

5.2 Why the vacancy formula and binomial factor have this form

Define the truncated string-box count

$$E_s^{(a)} = \sum_{s' \geq 1} \min(s, s') n_{s'}^{(a)}.$$

If the strings at level a are drawn as rows of lengths s', this counts boxes in their first s columns. For the ordinary type-A Cartan matrix, $A_{ab} = 2\delta_{ab} - \delta_{a,b+1} - \delta_{a,b-1}$, Eq. (35) becomes

$$P_s^{(a)} = \delta_{a,1} L - 2E_s^{(a)} + E_s^{(a-1)} + E_s^{(a+1)}, \qquad E_s^{(0)} = E_s^{(\ell)} = 0. \tag{37}$$

The neighbors contribute with plus signs after contracting the negative Cartan off-diagonal entries. The configuration must also satisfy the root-number constraint $\sum_s s\, n_s^{(a)} = M(a, 0)$. For each occupied string length, admissible highest-weight riggings require $P_s^{(a)} \geq 0$.

For n identical length-s strings and vacancy P, their labels form a weakly ordered list $0 \leq J_1 \leq \cdots \leq J_n \leq P$; exchanging equal-length strings creates no new configuration. The transformation $K_j = J_j + j$ gives a strictly increasing n-subset of $\{1, \ldots, P + n\}$. There are therefore $\binom{P+n}{n}$ possible riggings. Multiplying over levels and lengths gives Eq. (36); summing over all admissible string partitions is essential for recovering a shape's full count. A single string content does not generally account for all SYT.

Example 5.2 (The full rigging count for $(3, 2, 1)$)**.** *Here $(M_1, M_2) = (3, 1)$, so the level-two string partition is necessarily (1). The level-one possibilities are (3), $(2, 1)$, and $(1, 1, 1)$. The vacancy and rigging counts are*

Level-one partition	*Occupied level-one vacancies*	$P_1^{(2)}$	*Count*
(3)	$P_3^{(1)} = 1$	-1	*inadmissible*
$(2, 1)$	$P_1^{(1)} = 3$, $P_2^{(1)} = 1$	0	$4 \cdot 2 \cdot 1 = 8$
$(1, 1, 1)$	$P_1^{(1)} = 1$	1	$\binom{4}{3} \cdot 2 = 8$

For $(2, 1)$, for example, $E_1^{(1)} = 2$, $E_2^{(1)} = 3$, $E_1^{(2)} = E_2^{(2)} = 1$. Thus $P_1^{(1)} = 6 - 4 + 1 = 3$, $P_2^{(1)} = 6 - 6 + 1 = 1$, and $P_1^{(2)} = 2 - 2 = 0$. For (3) the second-level vacancy is $1 - 2 = -1$, so it is excluded even though its own level-one vacancy is positive. Summing $8 + 8 = 16$ agrees with the hook-length count. This counts the shape's combinatorial labels without assuming exact finite-size roots are perfect strings.

5.3 The Kerov–Kirillov–Reshetikhin bijection

The **Kerov–Kirillov–Reshetikhin (KKR) bijection** is a classical, constructive correspondence between rigged configurations and standard Young tableaux (equivalently, crystal paths in a tensor product of single-box representations). The project implements both directions of this bijection explicitly, following the classical algorithm: recursively remove (or add) the box associated with the current largest (or next) tableau entry, selecting eligible *singular* strings — those whose riggings equal their vacancy numbers — with the length constraints and tie-breaking rules of the algorithm, and updating vacancy numbers via (35) after each such move. The Wronskian solver does not

require this string construction. The separate string solver converts a rigged configuration into explicit **mode numbers** for each string (via a further shift of the rigging by the vacancy number and the string index) which, through the string ansatz (34), give an explicit approximate (string-hypothesis) prediction for the Bethe roots associated with a tableau, independent of the exact Wronskian construction of Sections 2–3.

5.4 A flip involution and a conjectured root symmetry

Complementing every string's rigging against its own vacancy number,

$$J \;\longmapsto\; P_s^{(a)} - J, \tag{38}$$

applied simultaneously to every string of a rigged configuration, defines an involution on rigged configurations (and, via the KKR bijection, an involution on standard Young tableaux of a fixed shape), which we denote SYT $\mapsto$ Flip(SYT).

Empirical Symmetry 5.1 (Root negation under the flip)**.** *For a standard Young tableau* SYT *with Bethe roots* $\{u_1, \ldots, u_r\}$ *(at any fixed nesting level), the flipped tableau's Bethe roots satisfy*

$$\{u_1, \ldots, u_r\} \;=\; \{-u_1, \ldots, -u_r\} \quad \textit{as computed for } \mathrm{Flip(SYT)}. \tag{39}$$

This statement is not derived in the codebase from first principles; it is used purely as an *empirical, numerically checked* consistency condition (Section 10.2). Combinatorially, the flip (38) coincides, on standard Young tableaux, with **Schützenberger evacuation** — the classical involution obtained by repeatedly removing the box containing the tableau's smallest entry via jeu-de-taquin slides and relabeling — a fact from crystal-base theory (the relevant crystal-path convention identifies the two involutions). The project exploits this coincidence purely as a fast, dependency-free way to compute the flipped tableau without constructing its rigged configuration explicitly, and treats Conjecture 5.1 as an independent numerical cross-check on the Wronskian-continuation output of Sections 2–4 rather than as a derivation of it.

6 Numerical Continuation: Formalism and Realization

6.1 A concrete warm-up example

Before setting up the general formalism, it is worth fixing ideas with the simplest possible instance, reproduced directly from the project's own introductory material. Take the target system

$$F(x, y) \;=\; \begin{pmatrix} x^2 + y^2 - 1 \\ x + y \end{pmatrix} \;=\; 0, \tag{40}$$

i.e. the intersection of the unit circle with the line $y = -x$, with exact solutions $(x, y) = (\pm\frac{1}{\sqrt{2}}, \mp\frac{1}{\sqrt{2}})$. This system has no free parameter to continue in, so one instead introduces an artificial *start system* $G(x, y) = 0$ of the same degrees (2 and 1, respectively) with known solutions — a **total-degree start system** — and a homotopy $H = (1 - t)F + tG$ exactly as in (41) below. Bézout's theorem bounds the number of solution paths that must be tracked by the product of the two degrees, $2 \times 1 = 2$; correspondingly there are two start points to track, one per path, from $t = 1$ down to $t = 0$. Tracking both paths with an adaptive multiple-precision corrector (as described below) recovers the two exact intersection points to the tracker's tolerance, with real parts $\approx \pm 0.70710678\ldots = \pm 1/\sqrt{2}$ — this is exactly what one finds running Bertini on (40), and it is the same mechanism, applied to

a genuine polynomial system with no natural deformation parameter of its own, that Section 6.3 specializes to the project's actual problem, where instead of an artificial start system one already has a refined solution at $\Lambda = \Lambda_0$ to continue from.

6.2 Homotopy continuation and the Davidenko equation

Let $F : \mathbb{C}^N \to \mathbb{C}^N$ be the target polynomial system (here, the Wronskian relations (26) at $\Lambda = 0$, in the coefficients γ of Section 3.4), and let G be a *start system* with known solution(s), continuously connected to F by a homotopy

$$H(x,t) \;=\; (1-t)\,F(x) + t\,G(x), \qquad t \in [0,1], \tag{41}$$

so that $H(x,1) = G(x)$ and $H(x,0) = F(x)$. Assuming the solution path $x(t)$ defined implicitly by $H(x(t),t) = 0$ is smooth, differentiating with respect to t gives the **Davidenko differential equation**,

$$J_x H\big(x(t),t\big)\,\dot{x}(t) \;+\; \partial_t H\big(x(t),t\big) \;=\; 0 \quad\Longrightarrow\quad \dot{x}(t) \;=\; -\big[J_x H(x(t),t)\big]^{-1}\,\partial_t H\big(x(t),t\big), \tag{42}$$

an ordinary differential equation for the path $x(t)$, provided the Jacobian $J_x H$ remains nonsingular along the path. Numerically tracking the path from $t = 1$ to $t = 0$ then reduces to (i) a **predictor** step that numerically integrates (42) by one of the standard explicit ODE schemes, and (ii) a **corrector** step that removes the resulting drift off the true solution path by Newton iteration,

$$x_{k+1} \;=\; x_k \;-\; \big[J_x H(x_k,t)\big]^{-1} H(x_k,t), \tag{43}$$

applied at fixed t until convergence to a specified tolerance. This predictor–corrector scheme, together with *adaptive* control of both the step size in t and the working numerical precision (raised whenever the Jacobian in (43) becomes ill-conditioned, as happens near a path singularity), is exactly what Bertini's **adaptive multiple-precision tracker** ("AMPTracker") implements; the several named predictors available (constant, Euler, Heun, and a family of embedded Runge–Kutta schemes of increasing order) are simply different choices of explicit ODE integrator for (42), trading predictor accuracy (and hence achievable step size) against per-step cost.

6.3 Specialization: a linear parameter homotopy in Λ

The construction of Section 4 is a particularly favorable instance of (41): rather than requiring an artificial start system, the Wronskian relations (26), left symbolic in Λ, already constitute a one-parameter polynomial family $F_\Lambda(\gamma) = 0$ whose member at a large value $\Lambda = \Lambda_0$ is solved by refining the combinatorial seed of Section 4, and whose member at $\Lambda = 0$ is the physical system. Introducing the affine reparametrization

$$\Lambda(t) \;=\; (1-t)\cdot 0 + t\cdot\Lambda_0 \;=\; t\,\Lambda_0, \qquad t \in [0,1], \tag{44}$$

gives $H(\gamma,t) := F_{\Lambda(t)}(\gamma)$, with $H(\gamma,1) = F_{\Lambda_0}(\gamma)$ solved numerically to the start tolerance and $H(\gamma,0) = F_0(\gamma)$ the target; exactly one path is tracked (there is no Bézout-number multiplicity of start points to enumerate, since one refined start point has been selected), from $t = 1$ to $t = 0$. This is the parameter homotopy actually solved by the numerical back end: the polynomial system exported at each step of the sequential construction (Section 4.3) is $F_\Lambda(\gamma)$ with Λ left as the single homotopy variable and (44) substituted in symbolically before the system is handed to the tracker, together with the refined starting value $\gamma(\Lambda_0)$ from Section 4 as the $t = 1$ start point.

6.4 A predictor–corrector step, with actual numbers

There are three different derivatives to keep separate. The Jacobian $J_{ij} = \partial F_i/\partial\gamma_j$ differentiates with respect to unknown coefficients, F_Λ differentiates with respect to the external parameter, and the total derivative along a solution obeys

$$J\frac{d\gamma}{d\Lambda} = -F_\Lambda. \tag{45}$$

For Eq. (53), $F_\Lambda = (1,0,0,0,0,0)^{\mathsf{T}}$. For the tracker parameter t, multiply by $d\Lambda/dt = \Lambda_0$. In practice solve this linear system for the tangent; do not explicitly form a matrix inverse.

At (γ_j, Λ_j), an Euler predictor gives $\gamma_{\text{pred}} = \gamma_j + \delta\Lambda\, v_j$ with $J_j v_j = -F_\Lambda$. Newton correction at the new, fixed parameter is

$$J(\gamma^{(k)}, \Lambda_{j+1})\,\delta\gamma^{(k)} = -F(\gamma^{(k)}, \Lambda_{j+1}), \qquad \gamma^{(k+1)} = \gamma^{(k)} + \delta\gamma^{(k)}.$$

The predictor estimates a nearby point on the same path; the corrector solves the new fiber. Newton convergence by itself does not identify which branch was found.

Use the exact three-site example to see the numbers. Its physical root obeys $g(r,\Lambda) = r^2 - 2\Lambda r/3 - 1/12 = 0$ and

$$\frac{dr}{d\Lambda} = \frac{2r/3}{2r - 2\Lambda/3} = \frac{1}{3}\left(1 + \frac{\Lambda}{\sqrt{\Lambda^2 + 3/4}}\right)$$

on the positive branch. Starting at $\Lambda = 1$, the root is $r = 0.774291885177\ldots$. Taking $\delta\Lambda = -0.1$ gives $r_{\text{pred}} = 0.715760920310\ldots$. One Newton update at $\Lambda = 0.9$ is

$$r_{\text{new}} = r_{\text{pred}} - \frac{r_{\text{pred}}^2 - 0.6 r_{\text{pred}} - 1/12}{2r_{\text{pred}} - 0.6} = 0.716333593754\ldots,$$

while the exact new root is $0.716333199893\ldots$. Further corrections remove the remaining error. The only approximation in this illustration is the finite predictor step and its stopped Newton correction; the polynomial family is exact.

6.5 Residuals, conditioning, and branch identity

A residual $\|F(\widehat{\gamma}, \Lambda)\|$ measures equation error. For a nearby regular solution, its linearized coefficient error is $\delta\gamma \simeq -J^{-1}F$. Hence a tiny residual can coexist with a larger coefficient error if J is ill-conditioned. Rescaling individual equations changes the numerical residual norm, so compare residuals in a stated normalization; it does not change the zero set. A practical endpoint check combines the following independent questions:

1. Are all coefficients finite, and is the exported real representation consistent within the requested tolerance?
2. Does the monic vacuum determinant match u^L coefficient by coefficient, and do reconstructed QQ relations have small residuals?
3. Do polynomial divisions have vanishing remainders and the prescribed degrees, including boundary values?
4. Are the roots grouped correctly by nesting level, and do the nonsingular nested BAE or their polynomial forms hold?

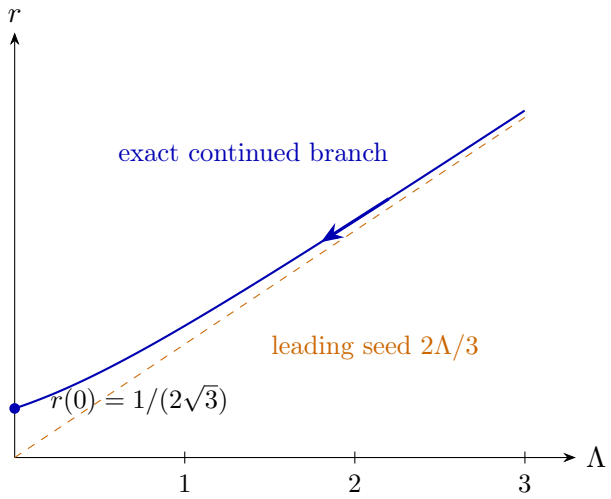


Figure 4: The asymptotic seed approaches the exact three-site branch only at large Λ. Continuation runs toward the physical endpoint on the left. Their difference near zero is not a rounding error.

5. Does continuation retain the intended tableau branch, as checked by reproducibility, reverse tracking, or comparison with a flip partner?

The final question is different from the first four: a different branch may solve the same endpoint equations perfectly. A flip check is useful additional evidence but cannot alone certify every tableau label.

6.6 Numerical safeguards

Two safeguards are imposed around the tracked endpoint before it is accepted as a physical solution. First, because the current interface exports a real representation of the tracked polynomial coefficients, each returned coefficient $\gamma = \gamma_{\mathrm{re}} + i\gamma_{\mathrm{im}}$ is required to satisfy

$$|\gamma_{\mathrm{im}}| \;\le\; \varepsilon_{\mathrm{track}} \cdot \max(1, |\gamma_{\mathrm{re}}|), \tag{46}$$

for the tracker's own tolerance $\varepsilon_{\mathrm{track}}$, with any violation (or any non-finite value) aborting the run rather than silently truncating an unexpectedly complex endpoint. Second, every numerical result is published to disk only by an atomic replace of a completed temporary file, so that a reader can never observe a partially written solution, and a failed or timed-out continuation simply leaves no new output rather than a corrupted one.

6.7 A path-tracking failure mode: branch jumps

The predictor–corrector scheme of (42)–(43) is only guaranteed to follow the correct solution branch if the step size in t is small enough, relative to the local curvature of the path and the separation of nearby solution branches, for the corrector to converge back to the *same* branch the predictor was following. When this fails — typically because the path passes close to another solution branch or to a singularity of the discriminant locus — the tracker can converge to a numerically valid but physically incorrect solution: a *branch jump*. Empirically, for a pair of tableaux related by the flip involution of Section 5.4 (whose roots are conjectured to be exact negatives of one another, Conjecture 5.1), starting the continuation from a substantially larger Λ_0 and tracking at higher

precision and tighter tolerance was found to eliminate an observed branch-jump discrepancy entirely (Section 10.2); a larger Λ_0 can improve the asymptotic seed but increases the parameter interval to be tracked. Higher precision reduces round-off, and tighter step control helps keep Newton correction on the intended branch. Neither change alone is a formal branch-identity guarantee.

Remark 6.1 (Why extra precision helps)**.** *It is worth making the mechanism a little more concrete. Two solution branches that pass close to one another correspond to the Jacobian $J_xH(x(t),t)$ in (43) becoming nearly singular somewhere along the path (formally, the path approaches the* discriminant locus *where two roots of the system coincide). Near such a point, the Newton correction $[J_xH]^{-1}H$ is the product of a huge, ill-conditioned inverse and a small residual, so the correction's accuracy is extremely sensitive to round-off in H and J_xH themselves: at fixed (say, machine) precision, the correction step can easily be thrown to the* wrong *one of the two nearby branches. Raising the working precision shrinks the round-off in H and J_xH directly, which is exactly what lets the corrector resolve which of the two nearby branches it started closest to — this is the sense in which* adaptive *precision (raised locally, only where the path actually passes near such a point, rather than uniformly everywhere) is doing real, not merely cosmetic, numerical work.*

7 A Fully Worked Example: the Shape $(2,1)$

The coefficient calculation in Eq. (9) already solves the complete Q-system. It is useful to recover the same roots from the ordinary one-magnon BAE, for a homogeneous length-three chain:

$$\left(\frac{u+i/2}{u-i/2}\right)^L = 1. \tag{47}$$

At $L=3$, the difference of cubes is

$$(u+i/2)^3-(u-i/2)^3 = i(3u^2-1/4)=0, \qquad u=\pm\frac{1}{2\sqrt{3}}. \tag{48}$$

Neither root makes the denominator zero. The two solutions agree with the roots $-x$ of $Q_{1,0}=u+x$ found from QQ coefficient matching. Equation (31) fixes T_A to the positive root for the specified positive-real inhomogeneity path; T_B gives the other branch. The shape alone determines the pair, while the path determines the assignment.

Evacuation exchanges T_A and T_B. To check it explicitly, delete 1 from T_A, slide 2 from the right into the hole (it is smaller than the lower neighbor 3), and record 3 at exit box $(1,2)$. Decrement the remaining entries. The next deletion exits at $(2,1)$, where one records 2; the last exits at $(1,1)$, where one records 1. The result is T_B. Their root sets are exact negatives, so the level-wise flip comparison vanishes. This is a complete warm-up with one root; the next section treats coupled nesting levels and a nonreal root pair.

8 A Nontrivial Worked Solution: One Tableau of Shape $(3,2,1)$

8.1 Choose a tableau, rather than only a shape

We solve the branch labeled by

$$T = \begin{array}{ccc} 1 & 2 & 6 \\ 3 & 4 & \\ 5 & & \end{array} = \{\{1,2,6\},\{3,4\},\{5\}\}. \tag{49}$$

All row and column inequalities hold. Its shape is self-conjugate and has six boxes. The hook lengths are

$$\begin{array}{ccc} 5 & 3 & 1 \\ 3 & 1 & \\ 1 & & \end{array}, \qquad f^{(3,2,1)} = \frac{6!}{5\cdot 3\cdot 1\cdot 3\cdot 1\cdot 1} = 16.$$

Thus there are sixteen tableau labels for the shape, not one. The homogeneous equations specify the shape; the growth sequence and continuation select the solution belonging to T.

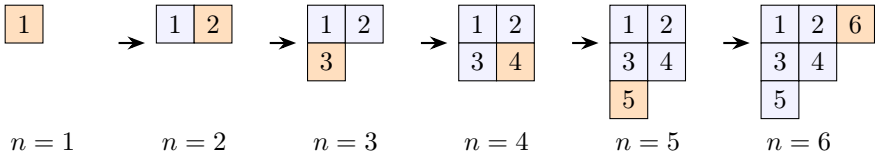


Figure 5: The growth sequence of the chosen tableau. Orange marks the newly added box. Steps $n = 1, 2$ are the trivial initial first-row segment.

8.2 Count degrees and write every unknown

The positive-degree polynomials are $Q_{0,0}$ of degree six, $Q_{1,0}, Q_{0,1}$ of degree three, and $Q_{2,0}, Q_{1,1}, Q_{0,2}$ of degree one. The required zero-degree boundary functions are $Q_{3,0} = Q_{3,1} = Q_{2,1} = Q_{2,2} = Q_{1,2} = Q_{1,3} = Q_{0,3} = 1$. The physical nesting levels contain $M_1 = 3$ and $M_2 = 1$ roots.

Use the default purely bosonic marked point $(3, 0)$. From Eq. (13), take gauge-fixed elementary polynomials

$$B_1 = u^5 + au^4 + bu^2 + c, \qquad B_2 = u^3 + du^2 + e, \qquad B_3 = u + f. \tag{50}$$

The omitted u^3, u terms of B_1, and u term of B_2, are gauge zeros. There are six unknowns (a, b, c, d, e, f). The normalized vacuum determinant is

$$Q^{\mathrm{W}}_{0,0}(u) = \frac{1}{-16i} \det \begin{pmatrix} B_1(u+i) & B_1(u) & B_1(u-i) \\ B_2(u+i) & B_2(u) & B_2(u-i) \\ B_3(u+i) & B_3(u) & B_3(u-i) \end{pmatrix}. \tag{51}$$

The leading coefficient is $i^3(5-3)(5-1)(3-1) = -16i$, explaining the denominator. Expand as $u^6 + w_5u^5 + \cdots + w_0$. Every coefficient is

$$\begin{aligned} w_5 &= \frac{3}{8}a + \frac{3}{4}d + \frac{15}{8}f, \\ w_4 &= \frac{3}{8}ad + \frac{3}{4}af + \frac{15}{8}df + \frac{5}{4}, \\ w_3 &= adf + \frac{1}{2}a - \frac{1}{8}b + \frac{5}{8}d - \frac{5}{4}e + \frac{5}{8}f, \\ w_2 &= \frac{3}{8}ad - \frac{3}{4}ae + \frac{3}{8}af - \frac{3}{8}bf + \frac{1}{4}, \\ w_1 &= \frac{1}{4}adf + \frac{1}{8}a - \frac{1}{8}b + \frac{3}{8}c - \frac{1}{8}d + \frac{5}{8}e + \frac{1}{4}f, \\ w_0 &= \frac{1}{8}(ae + af - be - bf + cd - df). \end{aligned} \tag{52}$$

At the final growth step, this is the square parameter system

$$F_6(a,b,c,d,e,f;\Lambda) = (w_5+\Lambda, w_4, w_3, w_2, w_1, w_0) = 0. \tag{53}$$

Its homogeneous endpoint is $w_k = 0$ for every k. No root variable is tracked in this system; roots are extracted afterward.

For reference, the two physical polynomials follow immediately from minors:

$$\begin{aligned} Q_{2,0} &= u + f, \\ Q_{1,0} &= \frac{\mathrm{Wr}[B_2, B_3]}{2i} = u^3 + \frac{d+3f}{2}u^2 + \left(df + \frac{1}{4}\right)u + \frac{d}{8} - \frac{e}{2} - \frac{f}{8}. \end{aligned} \tag{54}$$

This is the two-basis conversion worked out explicitly, including its normalization and constant term.

8.3 Follow the branch through all nontrivial shapes

The growth sequence is

$$(1) \to (2) \to (2,1) \to (2,2) \to (2,2,1) \to (3,2,1).$$

At $n = 3$ the exact positive branch is Eq. (31). For later steps, Eq. (28) gives the following new-root weights. A dash means that the vertex gains no new root, not that its coefficients must remain constant during continuation.

n	New box (r_n, c_n)	$N_{1,0}^{(n)}$	$N_{2,0}^{(n)}$	Other positive-degree vertices gaining a root
3	$(2,1)$	$2/3$	–	none
4	$(2,2)$	$1/2$	–	$N_{0,1} = 1/2$, $N_{1,1} = 1/4$
5	$(3,1)$	$3/4$	$1/2$	none
6	$(1,3)$	–	–	$N_{0,1} = 4/5$, $N_{0,2} = 8/15$

The vacuum gains root Λ at every step. For example, at $n = 5$

$$N_{2,0}^{(5)} = \frac{2+3-1-1}{2+3-1-1+1}\,\frac{2+3-1-2}{2+3-1-2+1} = \frac{3}{4}\,\frac{2}{3} = \frac{1}{2}.$$

The first-column seeds are consequently

$$\begin{aligned} n=3: &\quad Q_{1,0}^{\mathrm{seed}} = u - 2\Lambda/3, \\ n=4: &\quad Q_{1,0}^{\mathrm{seed}} = (u-\Lambda/2)(u-1/(2\sqrt{3})), \\ n=5: &\quad Q_{1,0}^{\mathrm{seed}} = (u-3\Lambda/4)(u^2+1/4), \quad Q_{2,0}^{\mathrm{seed}} = u - \Lambda/2, \\ n=6: &\quad Q_{1,0}^{\mathrm{seed}} = Q_{1,0}^{(5)}(u;0), \quad Q_{2,0}^{\mathrm{seed}} = Q_{2,0}^{(5)}(u;0). \end{aligned}$$

The final step adds no first-column roots, yet changes their values, since the vacuum and the equations have changed.

For a reproducible independent calculation, the companion `verify_worked_example.py` constructs the pure-bosonic determinant at each shape, refines at $\Lambda_0 = 30$, and solves at successive Λ values separated by 0.05 down to zero, using 60-digit arithmetic and Newton correction. This illustrative calculation is independent of the production Bertini tracker. It gives:

n	First-level roots	Second-level roots
3	0.288675134595	no second level
4	$\pm 0.5i$	no second level
5	$0.539099643380,\ -0.196076665153 \pm 0.500217863030i$	0.293892626146
6	$0.219903574307,\ -0.432700399337 \pm 0.503065694765i$	0

The largest unscaled vacuum-coefficient residual at the final step of this run was below 2×10^{-50}. This measures the solved equations; it does not certify every displayed digit or rule out branch jumps in arbitrary runs. The final root sets also agree with the saved row for Eq. (49) in the saved file

`data/references/all_SYT_{3,2,1}.csv`

to 10^{-12}. That CSV is saved-run evidence; the calculation here is not presented as a fresh execution of the Wolfram/Bertini production pipeline.

The $n = 4$ endpoint has singular ordinary BAE roots $\pm i/2$. The coefficient solution $Q_{1,0} = u^2 + 1/4$ remains a regular polynomial object. This intermediate step is a concrete reason to continue Q-function coefficients and avoid dividing root equations by vanishing factors.

8.4 Solve the selected homogeneous branch exactly

The branch above ends at $f = 0$, so $Q_{2,0} = u$. Do not impose $f = 0$ when solving all sixteen tableau branches: here it is a property of this selected endpoint. With $f = 0$, the six equations Eq. (53) become easy to eliminate:

1. $w_5 = 0$ gives $a = -2d$.
2. $w_4 = 0$ gives $3ad/8 + 5/4 = 0$, hence $d^2 = 5/3$.
3. $w_2 = 0$ gives $3ad/8 - 3ae/4 + 1/4 = 0$, hence $ae = -4/3$ and $e = 2/(3d) = 2d/5$.
4. $w_3 = 0$ gives $b = 4a + 5d - 10e = -7d$.
5. $w_1 = 0$ gives $c = (-a + b + d - 5e)/3 = -2d$.
6. $w_0 = 0$ is then satisfied: $(a - b)e + cd = 5de - 2d^2 = 0$.

The continued solution has $d > 0$, so $d = \sqrt{15}/3$. Therefore

$$(a, b, c, d, e, f) = \left(-\frac{2\sqrt{15}}{3}, -\frac{7\sqrt{15}}{3}, -\frac{2\sqrt{15}}{3}, \frac{\sqrt{15}}{3}, \frac{2\sqrt{15}}{15}, 0 \right). \tag{55}$$

Substituting into Eq. (54) yields the exact physical polynomials

$$\boxed{Q_{1,0} = u^3 + \frac{\sqrt{15}}{6}u^2 + \frac{1}{4}u - \frac{\sqrt{15}}{40}, \qquad Q_{2,0} = u.} \tag{56}$$

Both coefficient vectors are real, while $Q_{1,0}$ has two nonreal zeros. The positive-sign choice was fixed by the branch calculation, not by the homogeneous equations alone.

8.5 Reconstruct the whole Q-system and check every cell

Let $A = \sqrt{15}/6$, so $A^2 = 5/12$. From the QQ cell $(1, 0)$,

$$\frac{\Delta Q_{1,0}}{3i} = u^2 + \frac{2A}{3}u = Q_{2,0}Q_{1,1}, \qquad Q_{1,1} = u + \frac{2A}{3}.$$

For the cell $(0, 0)$, direct expansion gives

$$\frac{\mathrm{Wr}[u^6, u + 2A/3]}{5i} = u^6 + \frac{4A}{5}u^5 - \frac{1}{4}u^4 - \frac{2A}{3}u^3 - \frac{9}{80}u^2 + \frac{A}{20}u + \frac{1}{320}. \tag{57}$$

Divide this polynomial by $Q_{1,0}$ using $A^2 = 5/12$; the remainder vanishes and the quotient is $Q_{0,1} = u^3 - Au^2/5 - 5u/12 - A/20$. Next,

$$\frac{\Delta Q_{0,1}}{3i} = u^2 - \frac{2A}{15}u - \frac{2}{9} = (u + 2A/3)(u - 4A/5),$$

so $Q_{0,2} = u - 4A/5$. We have now reconstructed every positive-degree polynomial:

$$\begin{array}{ll} Q_{0,0} = u^6, & Q_{1,0} = u^3 + Au^2 + u/4 - 3A/20, \\ Q_{0,1} = u^3 - Au^2/5 - 5u/12 - A/20, & Q_{2,0} = u, \\ Q_{1,1} = u + 2A/3, & Q_{0,2} = u - 4A/5. \end{array} \tag{58}$$

The remaining QQ cells $(2, 0), (1, 1), (0, 2)$ reduce to $\Delta(u + \text{constant}) = i$, with unit boundary factors. Thus the six box relations are all verified, including the nontrivial vacuum relation and both polynomial divisions.

One can also reconstruct the quotient by coefficient matching. Set $Q_{0,1} = u^3 + Du^2 + Eu + F$ and $Q_{1,0} = u^3 + Au^2 + u/4 + C$. Equating their product to Eq. (57) gives

$$\begin{aligned} A/5 + D &= 0, & AD + E + 1/2 &= 0, \\ AE + 2A/3 + C + D/4 + F &= 0, & AF + CD + E/4 + 9/80 &= 0, \\ -A/20 + CE + F/4 &= 0, & CF - 1/320 &= 0. \end{aligned}$$

Their solution on this branch is $D = -A/5$, $E = -5/12$, $C = -3A/20$, $F = -A/20$. This provides a second, entirely explicit route to the same lattice data.

8.6 Extract the cubic roots, with no unspecified solver step

To solve the first polynomial in Eq. (56), put $u = z - A/3$. Since $A^2 = 5/12$, its depressed cubic is

$$z^3 + \frac{1}{9}z - \frac{82A}{405} = 0.$$

Its Cardano discriminant is

$$\left(\frac{41A}{405}\right)^2 + \left(\frac{1}{27}\right)^3 = \frac{7}{1620} > 0.$$

Let $\sqrt[3]{x}_{\mathbb{R}}$ denote the real cube root, also for negative x. The real Bethe root is exactly

$$r = \sqrt[3]{\frac{41\sqrt{15}}{2430} + \frac{\sqrt{35}}{90}}_{\mathbb{R}} + \sqrt[3]{\frac{41\sqrt{15}}{2430} - \frac{\sqrt{35}}{90}}_{\mathbb{R}} - \frac{\sqrt{15}}{18}. \tag{59}$$

Dividing by $u - r$ leaves $u^2 + (A + r)u + (1/4 + Ar + r^2)$, so the other two roots are

$$u_{\pm} = -\frac{A + r}{2} \pm \frac{i}{2}\sqrt{1 + 2Ar + 3r^2 - A^2}. \tag{60}$$

Numerically, the root lists by nesting level are

$$\{u_j^{(1)}\} = \{0.2199035743065745, -0.4327003993372387 \pm 0.5030656947652822i\},$$
$$\{u_j^{(2)}\} = \{0\}. \tag{61}$$

These decimals are rounded displays of the exact solution, not additional independent unknowns.

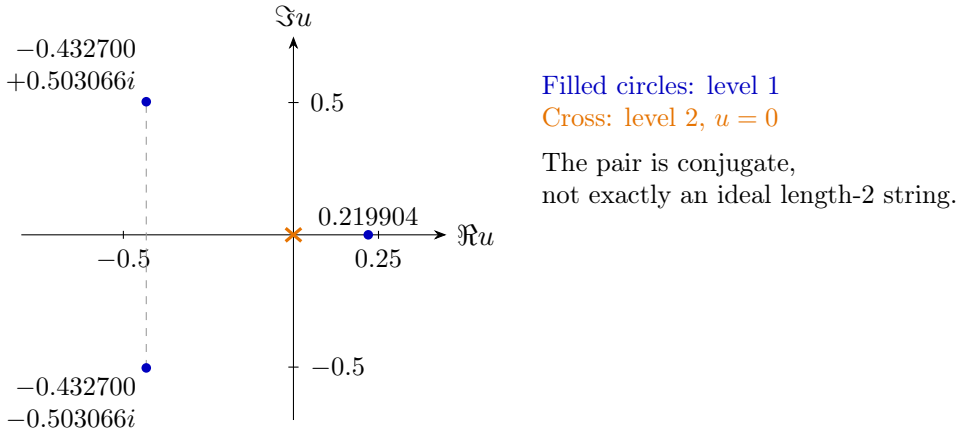


Figure 6: Complex rapidity plane for the exact six-box solution. The second-level zero is retained as a separate root family.

As an independent root-equation check, insert these polynomials into Eq. (33). At level two, $v = 0$ and $Q_3 = 1$; $Q_1(i/2) = Q_1(-i/2) = -2A/5$, so the left side is 1. The right side is $-Q_2(i)/Q_2(-i) = 1$ as well. At level one the pole-cancellation numerator reduces modulo Q_1 to zero; the companion script checks this exactly, so the cubic roots all satisfy the ordinary BAE wherever the shifted denominators are nonzero.

8.7 Strings and evacuation: what this example validates

The complex pair has center $v \approx -0.432700399337$. Its imaginary separation is approximately 1.006131389531. An ideal length-two string would have roots $v \pm i/2$. The observed deviation of each imaginary part is 0.0030656947652822. The exact polynomial solution therefore illustrates both the usefulness and the finite-size imperfection of the string picture; its rigged configuration is combinatorial data, not an assertion that its exact roots have spacing precisely i.

Evacuation gives

$$\mathrm{Flip}(T) = \begin{array}{ccc} 1 & 2 & 4 \\ 3 & 6 & \\ 5 & & \end{array}.$$

To reproduce it, delete the smallest remaining entry, slide the smaller of the right and lower neighbors into each hole, record $6, 5, \ldots, 1$ at successive exit boxes, and decrement the remaining entries after each exit. For this T, the exit positions in one-based box coordinates are

$$(2,2),\ (3,1),\ (1,3),\ (2,1),\ (1,2),\ (1,1).$$

The negated polynomial solution is

$$\widetilde{Q}_{a,s}(u) = (-1)^{M(a,s)} Q_{a,s}(-u).$$

Indeed, reflecting u swaps $+$ and $-$, while the degree balance in Eq. (6) supplies the compensating sign. The vacuum u^6 and unit boundary polynomials remain unchanged. This proves reflection invariance of the homogeneous equations. Matching the reflected solution to the evacuated tableau is the extra label statement checked empirically in the repository. The companion script independently continues the evacuated tableau and compares its roots with the negatives of Eq. (61) level by level at tolerance 10^{-12}. The checked reference CSV contains T but does not contain this flip partner; the partner check is therefore an independent numerical calculation, not a saved-reference comparison.

9 Software Realization

This section summarizes, at the level of architecture rather than implementation detail, how the mathematical and continuation formalism is realized as a working pipeline; it is included for completeness and to ground the validation results of Section 10, not as a substitute for the mathematical description above.

Symbolic layer (Wolfram Language). Given a Young diagram, a symbolic layer constructs the Q-system lattice, chooses a marked point (purely bosonic by default), builds the super-Wronskian expression (25), and derives the Wronskian relations (26) together with the two-basis translation of Section 3.4. A second, independent module implements the asymptotic construction of Section 4 and drives the sequential, box-by-box computation of Section 4.3 for a given standard Young tableau, delegating each individual continuation step to a pluggable numerical back end.

Numerical back end (Bertini, via Python). At each continuation step, the current Wronskian relations (with Λ left symbolic per (44)) and a high-precision starting point are exported to a plain-text table (one polynomial equation per row, each row carrying the variable name, its starting value, and its defining expression). A Python driver parses this table into a genuine polynomial system, forms the parameter path of (44), and tracks a single path with Bertini's adaptive multiple-precision tracker (Section 6), exposing the tracker's tolerance, precision, step-size, Newton-iteration, and predictor-scheme parameters as configurable settings. The converged coefficients are validated against the reality condition (46) and published atomically; the symbolic layer imports them and proceeds to the next site. *(An earlier version of this back end delegated the same step to a Julia-based homotopy-continuation package; several internal option and file names still reflect that earlier design, though the currently used back end is Bertini only.)*

Orchestration and batch execution. A thin orchestration layer isolates every numerical global used by the symbolic/asymptotic construction on a per-task basis, allocates a dedicated working directory and log files per tableau, and records detailed per-stage timing. A shell-level batch launcher runs many tableaux (optionally all standard Young tableaux of a given shape) as independent, parallel operating-system processes, consolidating their individual results into a single aggregate table under a file lock, atomically, so that a batch's partial results remain safely readable while it is still running; a native desktop application provides the same set of operations (single-tableau runs, batch runs, the flip cross-check of Section 10.2, and result visualization) through a graphical interface, without introducing any code path distinct from the command-line tools.

Visualization. Bethe roots are visualized either in a notebook (for interactive inspection and for the negated-overlay comparison used in the flip check) or by a small, dependency-free tool that renders

each nesting level of (32) as a panel of a hand-written vector-graphics document, automatically switching to a density-aggregated rendering when the number of roots in a batch result is too large to plot individually.

Remark 9.1 (Why split the pipeline this way)**.** *The division between a symbolic layer and a numerical back end mirrors the division between exact symbolic system construction and numerical solution. Combinatorial seed construction is exact arithmetic, but finite-start refinement already involves numerical approximation; Bertini then performs the remaining path tracking. Consequently, the only interface the two layers need is the plain-text polynomial table described above: the symbolic layer never needs to know* how *a step's solution was obtained, only that it converged, which is precisely what allowed the back end to be replaced (Section 9's remark on the earlier Julia back end) without touching the mathematics of Sections 2–4 at all.*

10 Validation and Numerical Results

10.1 Regression testing

The numeric-string handling between Bertini's long-decimal output and the symbolic layer's high-precision numbers is directly tested: a 100-digit decimal value is required to round-trip through export and import with every digit intact, standard and scientific notations are required to parse correctly, and malformed or non-finite input is required to be rejected rather than silently coerced. Separately, a suite of saved reference tableaux is rerun from scratch and compared against their saved solutions *level by level* (per Eq. (32), never flattened across nesting levels), using an exhaustive one-to-one matching over root permutations at a default tolerance of 10^{-8}.

10.2 Validation of Conjecture 5.1: the flip check

The empirical symmetry of Conjecture 5.1 is checked numerically by computing, for a tableau and its flip, the **symmetric Hausdorff distance** between one root set and the negation of the other,

$$d_H(A,B) \;=\; \max\Big\{ \max_{a\in A}\min_{b\in B}|a-b|,\;\; \max_{b\in B}\min_{a\in A}|a-b| \Big\}, \qquad B = -\,\mathrm{Roots}[\mathrm{Flip}(\mathrm{SYT})], \tag{62}$$

computed at high (100-digit) arithmetic precision, with a default pass tolerance of 10^{-5}; the check is applied automatically, at the level of a whole aggregate batch of tableaux, and any pair exceeding tolerance can be flagged and removed as a likely branch-jump artifact (Section 6.7). This is exactly the automated, large-scale counterpart of the by-hand check carried out in Section 7: there, d_H applied to $\{+\frac{1}{2\sqrt{3}}\}$ and $-\{-\frac{1}{2\sqrt{3}}\}$ vanishes identically, since the two single-element sets are literally equal; for shapes too large to solve in closed form, the same comparison is instead carried out numerically, at whatever precision the continuation was run at, on every flip-related pair in a batch.

Young diagram	Tableau	Max. root difference
$\{3,2,1\}$	$\{\{1,2,6\},\{3,4\},\{5\}\}$	4.7×10^{-17}
$\{3,2,1\}$	$\{\{1,2,6\},\{3,5\},\{4\}\}$	2.9×10^{-17}
$\{4,2,1\}$	$\{\{1,2,3,4\},\{5,6\},\{7\}\}$	1.4×10^{-14}

Table 2: Maximum one-to-one root differences (level by level) between freshly recomputed and saved reference solutions, all well below the 10^{-8} pass tolerance used for the comparison.

Table 2 reports a representative set of from-scratch reruns compared against saved reference solutions; all three cases agree to well within tolerance. This establishes agreement with previously computed (and themselves once machine-precision-validated) references to the stated tolerance; it does not, by itself, certify absolute accuracy at the full working precision used internally.

10.3 Performance profile of a single tableau

Table 3 reports a non-overlapping timing breakdown for one complete computation of a seven-box tableau, totaling 8.95 s, with recovered roots matching a saved reference to within 6.6×10^{-17}.

Category	Seconds	Share
Symbolic (Wolfram Language) construction & equation preparation	3.43	38.3%
Bertini path tracking	0.92	10.3%
Other back-end process overhead	1.04	11.6%
Kernel launch/load/finalization overhead	3.57	39.9%

Table 3: Timing breakdown for one small tableau. Bertini's own path-tracking cost is a modest fraction of the total; fixed per-call kernel and symbolic-preparation overhead dominates at this problem size.

For this particular (small) example, fixed overhead — Wolfram Language kernel initialization and one-time symbolic-preparation cost — dominates the wall-clock time, while the genuine numerical continuation cost (Bertini path tracking) is comparatively small; this conclusion is specific to the profiled example and has not been established for substantially larger Young diagrams, where the polynomial degrees, and hence the intrinsic path-tracking cost, grow.

11 Discussion

11.1 Summary

The practical unknowns are gauge-fixed elementary polynomial coefficients. Young-diagram geometry supplies their degrees and the monic QQ prefactors. Casoratian determinants and mixed discrete integration supply the Wronskian equations. A tableau specifies the sequence of large-root seeds; finite-start refinement and parameter continuation solve each nontrivial growth step. Reconstructed lattice polynomials yield roots separately at each nesting level.

The two worked examples make the distinction between shape equations and tableau branch labels concrete. For $(3, 2, 1)$, six elementary coefficients determine all six positive-degree Q-functions; one selected branch has an exact cubic and a second-level root at zero. All QQ cells, the nested root equations, the sixteen-label combinatorial count, and comparison with saved roots can be checked independently. The illustrative numerical growth calculation and exact algebra checks are reproducible with the companion script.

11.2 Limitations

This report uses the ordinary first-column physical interpretation; it does not infer a particular supersymmetric Hamiltonian solely from an auxiliary marked-point choice. The flip symmetry of Conjecture 5.1 is used, and validated numerically, but is not derived here from the representation

theory of the underlying algebra. The branch-jump failure mode of Section 6.7 is mitigated by more conservative continuation settings but is not eliminated by any formal guarantee, and should continue to be screened for via the flip check on any newly computed batch of tableaux. Performance characterization to date covers only small tableaux; whether fixed overhead continues to dominate genuine path-tracking cost at large Young-diagram size remains open.

11.3 Outlook

Two directions follow directly from the results above. First, since the dominant cost observed in Table 3 is fixed per-call overhead rather than path-tracking time, reusing a single initialized symbolic-layer worker across many tableaux (rather than paying kernel startup once per tableau) is a natural, low-risk optimization for batch computations. Second, extending the performance profile of Section 10 to substantially larger Young diagrams would determine whether the numerical continuation itself, rather than fixed overhead, becomes the dominant cost once the Q-system's polynomial degrees grow, which would in turn indicate where further numerical effort (tighter path-tracking control, higher base precision, or an adaptive step-size schedule tuned specifically to this problem's observed branch-jump risk) is best directed.

References

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[2] C. Marboe and D. Volin, *Fast analytic solver of rational Bethe equations*, arXiv:1608.06504, version 3 (2017).